Technical and Engineering Documentation

Drone Swarm Control and Intelligent Agricultural Protection

Module A: Engineering Design and Development

ITEM	DETAILS
DOCUMENT VERSION	Version 1
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PROJECT MANAGER	Sobhan Memarzadeh
OVERALL IMPLEMENTATION AND FINAL APPROVAL	Sobhan Memarzadeh
DOCUMENT STATUS	Proposed Design / Under Development

1. Project Overview and Objectives

1.1. Project Title

Design and Development of a Coordinated Three-Drone System for Intelligent Agricultural Monitoring, Identification of Areas Suspected of Pest Infestation, and Specialized Mission Management.

1.2. Problem Statement

Continuous monitoring of orchards and large agricultural areas requires time, personnel, and adequate access to different parts of the site. Ground inspections may face coverage limitations and delays in identifying signs of infestation or thermal hotspots. Using a single drone also limits the ability to conduct different missions simultaneously.

The Shahid Mohajer team's project aims to design a three-drone system that uses a ground control station, data exchange, and mission allocation to support coordinated monitoring and follow-up inspections.

Software development follows two parallel tracks: improving the user interface and user experience of the existing **skyris.pro** platform, and developing a basic, independent application as a foundation for the progressive implementation of coordinated multi-drone capabilities.

1.3. Main Objectives

- Design centralized management for a three-drone fleet.
- Plan flight paths and allocate mission areas.
- Improve information presentation and operator interaction.
- Develop an independent software platform for implementing swarming capabilities.
- Establish the infrastructure for video and telemetry display.
- Progressively develop agricultural image analysis.
- Design thermal-anomaly detection and alerting workflows.
- Design mission assignment workflows.
- Investigate the use of ArUco markers for pose analysis.
- Provide conceptual designs and a graphical prototype for modular payload management.

1.4. Project Component Status

AREA	STATUS
UI/UX ENHANCEMENT OF THE SKYRIS.PRO PLATFORM	User interface and user experience enhancements completed
TEAM-DEVELOPED INDEPENDENT SOFTWARE	Basic version under development
SWARMING CAPABILITY IN THE INDEPENDENT SOFTWARE	Planned for development and validation
COORDINATED MANAGEMENT OF THREE DRONES	In the design and development stage
IMAGE ANALYSIS AND PEST DETECTION	Requires development and evaluation using valid data
THERMAL-ANOMALY DETECTION	Planned for development
ARUCO ANALYSIS TOOL	Part of the development plan
MODULAR PAYLOADS	Concept and graphical prototype only; no physical implementation

2. Team Structure and Responsibilities

2.1. Team Member Roles

TEAM MEMBER	ROLE	RESPONSIBILITIES
SOBHAN MEMARZADEH	Project Manager; Overall Implementation and Final Approval	Development management, subsystem coordination, decision-making, and final approval of activities
HOSSEIN SHIRANI	User Interface Lead and Coordinator for Safety-Critical Activities	UI/UX enhancement; coordination of flight readiness, payload testing, and changes to safety parameters
MEHDI AGHA ZIARATI	Network and Distributed Systems Lead	Communication design, message exchange, network services, and link-quality monitoring
ALI GHAYOURI	Flight Operations, Computer Vision, and Thermal Imaging Lead	Participation in flight preparation, image analysis, and sensor calibration

2.2. Responsibilities for Safety-Critical Activities

ACTIVITY	EXECUTION COORDINATION	OVERALL IMPLEMENTATION AND FINAL APPROVAL
FLIGHT-READINESS REVIEW AND APPROVAL	Hossein Shirani	Sobhan Memarzadeh
PLANNING AND COORDINATION OF PAYLOAD TESTING	Hossein Shirani	Sobhan Memarzadeh
RECORDING AND COORDINATING CHANGES TO SAFETY PARAMETERS	Hossein Shirani	Sobhan Memarzadeh

Physical payload testing will begin only after a prototype has been built and the necessary technical and safety reviews have been completed.

3. Technical Description of the System

3.1. Path Planning and Autonomous Flight

In the proposed architecture, each drone's flight path is determined according to the operational area, altitude, speed, battery capacity, and inspection waypoints. Before mission execution, paths must be checked for boundary violations, interference with known obstacles, and potential conflicts between drones.

The positioning source is selected according to the operating environment. GNSS may be used outdoors, while methods such as visual positioning and ArUco-based positioning may be considered in suitable environments without GNSS availability.

Geographic coordinates, local coordinates, camera coordinates, and drone body coordinates must each have a defined reference frame, with explicit transformations between them.

3.2. Fleet Management Through the Graphical Interface

The fleet-management interface is designed to provide a clear view of drone status and reduce the complexity of operator interaction. Its main components include:

- Displaying each drone's identifier and role.
- Displaying battery, communication, and positioning status.
- Selecting an individual drone or a group of drones.
- Viewing flight paths and mission status.
- Displaying received video.
- Viewing alerts and events.
- Displaying command transmission, receipt, and execution status.

Controls without an operational connection must be identified as "Demonstration" or "Under Development." Group-command execution must also depend on independent readiness and safety checks for each drone.

3.3. Motion Coordination and Collision-Risk Management

The proposed motion-coordination design combines centralized planning with independent safeguards on each drone. The coordinator manages missions and high-level paths, while each drone checks the validity of its position estimate, the freshness of received data, and its communication status.

Permissible separation between drones must be determined with consideration of:

- Airframe dimensions and propeller swept areas.
- Positioning error.
- Relative closing speed.
- Communication and processing latency.
- Deceleration and stopping capability.
- Aerodynamic effects from neighboring drones.
- Operational safety margins.

At this stage, a fixed separation of 0.5 meters has not been validated as safe. Operational separation must be determined through analysis and testing appropriate to the drones' specifications.

If communication is lost or positioning becomes invalid, the fallback response must account for the environment, remaining energy, and the status of other drones. Holding position, returning, and landing are possible responses whose suitability must be evaluated separately.

3.4. Video and Telemetry Transmission

Each video stream and telemetry data stream must be associated with a drone identifier and a timestamp. Video traffic must be managed so that it does not interfere with command and safety-related messages.

The main requirements are:

- Adjusting video quality to available network capacity.
- Indicating disconnected or stale video.
- Distinguishing video-link status from control-link status.
- Checking telemetry data freshness.
- Measuring actual latency.
- Recording data for mission review where the infrastructure permits.

Metrics such as latency, frame rate, message loss, and communication range will be measured during practical testing.

3.5. Modular Payload System

The payload component has been developed at the level of **ideas, conceptual design, and a graphical prototype**. At present, no physical payload has been built, installed on a drone, or tested.

The purpose of this component is to design payload selection, status presentation, and user interaction with the relevant interface panel.

PROPOSED MODULE	CONCEPTUAL APPLICATION	SCOPE OF GRAPHICAL DESIGN	PHYSICAL STATUS
OPTICAL OR LASER MODULE	Illustrating a target-marking concept	Module selection and proposed status display	Not built or tested
PAYLOAD-RELEASE MODULE	Illustrating a payload-release concept	Request, confirmation, and status-display workflow	Not built or tested
SPRAYING MODULE	Illustrating a localized-treatment concept	Proposed controls and parameter displays	Not built or tested

Values displayed in the graphical prototype, such as tank capacity, remaining payload, or activation status, are illustrative until connected to actual hardware and sensors.

Future physical development of these modules will require assessment of mass, center of gravity, power consumption, mechanical and electrical interfaces, safety interlocks, and applicable legal requirements.

3.6. Thermal-Anomaly Detection

Thermal-anomaly detection is planned as a future capability. Its purpose is to identify suspicious areas and issue alerts for further inspection.

The system output must distinguish between three levels:

1. A thermal anomaly requiring investigation.
2. A suspected fire.
3. A fire confirmed through additional verification.

Thermal imagery must be interpreted in light of environmental conditions, surface characteristics, and unrelated heat sources. If the camera does not provide calibrated temperature output, reporting reliable absolute temperatures is outside the scope of this capability.

The proposed scope is detection and alerting; it does not include physical fire suppression.

3.7. Software Development Through Two Parallel Tracks

The Shahid Mohajer team is pursuing two parallel and complementary software-development tracks:

- UI/UX enhancement of the existing **skyris.pro** platform.
- Development of a basic, independent application for the progressive implementation of swarming.

3.7.1. Track One: UI/UX Enhancement of the skyris.pro Platform

In this track, the company's default software version serves as the starting point, and the team's work focuses on improving the user interface and user experience.

The objective is to improve information presentation, interface organization, and operator interaction with the software environment.

The completed work is limited to UI/UX. Development of the flight-control core, communication protocols, operational payload integration, and automated capabilities is outside the scope of this enhancement and is considered part of future work.

Any subsequent code development or operational integration with this platform must comply with lawful access, software licensing, and authorized interfaces.

3.7.2. Track Two: Development of the Team's Independent Software

Alongside the enhancement of the existing platform, the team began developing an independent application at a basic, early stage during the competition.

The purpose of this track is to establish an extensible platform under the team's management for progressively implementing fleet management and coordinated multi-drone control. As the project's second software platform, it enables swarming development to proceed in parallel.

Drone connectivity, data exchange, mission allocation, and motion coordination must be developed incrementally and evaluated first in simulation and then through controlled testing.

The basic software version is under development. Completion and operational validation of swarming remain objectives for subsequent development.

3.7.3. Relationship Between the Two Development Tracks

The two tracks complement one another. The first focuses on improving the user experience of an existing platform, while the second supports the development and testing of team-specific capabilities. Experience and feedback from interface design can also inform the independent application.

ASPECT	SKYRIS.PRO PLATFORM	TEAM'S INDEPENDENT SOFTWARE
DEVELOPMENT BASIS	Default version of the existing software	Basic version developed by the team
CURRENT FOCUS	UI/UX enhancement	Initial software infrastructure
STATUS	UI/UX enhancement completed	Under development
ROLE OF SWARMING	Outside the scope of the current UI/UX enhancement	A main objective of further development
OPERATIONAL CAPABILITY DEVELOPMENT	Depends on access, licensing, and authorized interfaces	Depends on infrastructure completion, integration, and testing
EVALUATION APPROACH	Interface and usability evaluation	Software testing, simulation, and controlled testing

3.8. ArUco Analysis Tool

The planned ArUco analysis tool is intended to inspect marker maps and support troubleshooting of pose estimation.

Its main inputs include:

- Dictionary type.
- Marker identifiers.
- Actual marker dimensions.
- Marker installation positions and orientations.
- Camera calibration parameters.
- Camera-to-drone-body transformation.

Proposed functions include displaying identifiers, checking marker-map consistency, visualizing coordinate axes, and examining reprojection error.

Geometric pose estimation with OpenCV depends on camera calibration parameters and actual marker dimensions. The tool will complement the calibration and positioning process. [1]

3.9. Pest Detection and Identification of Infestation Symptoms

The computer-vision task must be defined according to the available data and intended application.

TASK TYPE	OUTPUT	MAIN LIMITATION
DIRECT PEST DETECTION	Location of a visible pest in an image	Dependence on target size, image resolution, and pest visibility
DETECTION OF INFESTATION SYMPTOMS	A suspicious area or tree	Potential similarity to water stress, nutrient stress, or other diseases
PLANT-HEALTH CLASSIFICATION	Healthy, suspicious, or infested label	Need for reliable reference labels and diverse samples

For pests concealed within plant tissue, direct detection from aerial imagery is not always feasible. Research on red palm weevil and temporal UAV imagery provides a basis for investigating infestation symptoms, but practical application requires local data and evaluation. [2]

The model-development process includes:

1. Precisely defining the task and classes.
2. Collecting imagery representative of the intended operating conditions.
3. Labeling data with expert input.
4. Separating training, validation, and test data.
5. Training and comparing models.
6. Evaluating performance on the target hardware.
7. Analyzing errors and defining the application's operating limits.

To reduce data leakage, highly similar images from the same flight or tree must be handled carefully when dividing the dataset.

If object detection is selected as the task, an initial target may be mAP@0.5 ≥ 0.85 on an independent test set. This metric is not equivalent to overall accuracy and must be considered alongside Precision, Recall, mAP@0.5:0.95, and inference time. [3]

3.10. Mission Assignment

The proposed mission-assignment workflow is:

Target Detection → Event Registration → Validity and Safety Checks → Drone Selection → Receipt Acknowledgment → Follow-Up Inspection → Result Reporting

Drone selection considers distance, remaining energy, communication status, equipment, and current workload.

ALLOCATION CRITERION	ROLE IN THE DECISION
DISTANCE TO TARGET	Estimating travel time and cost
REMAINING ENERGY	Assessing mission feasibility and safe completion
COMMUNICATION STATUS	Ensuring the required data exchange is possible
EQUIPMENT	Matching drone capability to mission requirements
CURRENT MISSION STATUS	Preventing conflicting assignments
POSITION VALIDITY	Preventing decisions based on invalid data

Each mission must have a unique identifier, an expiration time, and a defined status to prevent duplicate or stale requests from triggering repeated execution.

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4. Project Significance and Review of Existing Solutions

4.1. Significance for Agriculture

According to FAO, up to 40 percent of global crop production is lost to plant pests and diseases each year. This highlights the global importance of plant monitoring and protection. [4]

The project's proposed value includes:

- More systematic coverage of agricultural land.
- Recording visual and thermal evidence.
- Identifying areas requiring inspection.
- Conducting complementary missions in parallel.
- Consolidating information at the ground control station.
- Supporting post-mission review.

Reduced pesticide use, inspection costs, or crop losses are anticipated benefits that require comparative testing to demonstrate.

4.2. Comparison of Solutions

SOLUTION	RELEVANT CAPABILITIES	CONSIDERATIONS RELATIVE TO THE PROJECT'S OBJECTIVES
QGROUNDCONTROL	Mission planning and multi-vehicle support	A dedicated pest-detection and agricultural-intervention workflow requires development and integration
MISSION PLANNER	Flight planning and management, with limited formation-flying capability	Its swarming functionality is described in the documentation as limited and experimental
SINGLE-DRONE SYSTEM	Monitoring with lower coordination complexity	More limited capacity for simultaneous independent missions
SHAHID MOHAJER TEAM'S PROPOSED SYSTEM	Proposed integration of fleet management, agricultural analysis, and user interface	Requires subsystem completion and performance validation

Both QGroundControl and Mission Planner offer capabilities related to multi-vehicle management. The project therefore seeks differentiation through agricultural workflow integration and a tailored user experience, rather than by claiming that existing multi-vehicle solutions are unavailable. [5][6]

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5. Proposed Architecture and Development Tools

5.1. Architectural Layers

LAYER	MAIN FUNCTION	DESIGN CONSIDERATION
USER INTERFACE	Displaying status and receiving operator requests	Distinguishing demonstrative controls from operational controls

MISSION COORDINATOR	Allocating tasks and managing mission states	Checking request validity, duplication, and expiration
COMPANION COMPUTING	Image analysis and high-level logic	Preserving the independence of basic flight-controller safeguards
FLIGHT CONTROLLER	Stabilization and execution of valid commands	Emergency behavior independent of the user interface
NETWORK	Transmitting commands, telemetry, and video	Monitoring link quality and data freshness
PAYLOAD	Performing specialized actions in future stages	Currently limited to conceptual and graphical design

Position data must include a timestamp, reference frame, and validity status. Coordinate transformations must also be explicitly defined and tested.

5.2. Proposed Tools

TOOL	PROPOSED APPLICATION	SELECTION CONSIDERATION
PX4	Flight control and integration with high-level processing	Hardware compatibility and emergency-response configuration
ROS 2	Data exchange between subsystems	Selecting QoS policies appropriate to each message type
OPENCV	Image processing, calibration, and ArUco	Output quality depends on calibration and imaging conditions
PYTORCH	Model training and evaluation	Performance must be assessed on the target hardware
QT / PYSIDE	Desktop-interface development	Compatibility with project technology and licensing requirements
GIT	Version control for code and documentation	Linking test reports to specific versions

These tools are proposed options for further development. In ROS 2, QoS policies must match the needs of the data; for example, the sensor-data profile commonly uses Best Effort reliability. Reliable delivery alone does not guarantee successful command execution at the application level. [7]

6. Safety and Risk Management

6.1. Safety Principles

- Development progresses from software prototypes and simulation to staged physical testing.
- Changes to safety-critical parameters must be recorded and approved.
- Demonstrative controls must be distinguishable from operational controls.
- Basic safeguards must not depend solely on the user interface or server.
- Emergency payload shutdown must be distinguished from flight-motor shutdown.
- Multi-drone takeoff, return, and landing sequences must be reviewed for path conflicts.
- Safety-critical actions must depend on valid conditions and authorization from the responsible person.

6.2. Main Risks

RISK	POTENTIAL CONSEQUENCE	PROPOSED MITIGATION
MISTAKING A DEMONSTRATIVE CONTROL FOR AN OPERATIONAL ONE	Incorrect understanding of system status	Clear labeling and disabling controls without an operational connection
COMMUNICATION LOSS	Loss of monitoring or high-level control	Independent, tested emergency behavior
STALE POSITION DATA	Incorrect decisions and increased proximity risk	Timestamping and data-freshness checks
COORDINATE-FRAME ERROR	Movement toward an incorrect location	Defined reference frames and transformation testing
INCORRECT PEST OR FIRE DETECTION	False alerts or missed events	Independent evaluation and human confirmation for safety-critical actions
UNINTENDED ACTIVATION OF A FUTURE PAYLOAD	Unwanted action or injury	Safety interlocks and a default inactive state

MESSAGE SPOOFING OR TAMPERING	Unauthorized commands	Authentication and access control
OPERATOR OVERLOAD	Delayed response or decision error	Alert prioritization and interface simplification

MAVLink 2 message signing supports message authentication but does not encrypt message content. Communication confidentiality requires a separate mechanism appropriate to the architecture. [8]

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7. Testing and Evaluation Plan

7.1. Testing Stages

STAGE	EVALUATION SCOPE	EXIT CRITERION
GRAPHICAL-PROTOTYPE EVALUATION	Status clarity, workflow, and control differentiation	Resolution of major user ambiguities
SOFTWARE TESTING	Messages, allocation, and error handling	Passing the defined scenarios
MULTI-DRONE SIMULATION	Paths, latency, communication loss, and mission conflicts	Predictable and traceable behavior
BENCH TESTING	Future hardware connections and commands	Safety approval before flight testing
SINGLE-DRONE FLIGHT	Control, communication, and emergency behavior	Readiness approval before increasing drone count
MULTI-DRONE TESTING	Coordination and separation management	Staged execution in a controlled environment

7.2. Evaluation Metrics

AREA	METRICS
USER INTERFACE	Task-completion time, user errors, and status clarity
PATH FOLLOWING	Path-tracking error and boundary violations
COORDINATION	Minimum recorded separation and violations of permitted separation

MISSION ALLOCATION	Allocation time, success rate, and duplicate requests
COMMUNICATION	Latency, message loss, and recovery time
VIDEO	Display latency, frame loss, and stream stability
COMPUTER VISION	Precision, Recall, and task-appropriate metrics
THERMAL DETECTION	False alerts, missed events, and alerting time
ARUCO	Detection rate, reprojection error, and position error against a reference
OPERATIONS	Preparation time, operator workload, and time to safe termination

To assess the benefits of using multiple drones, single-drone and three-drone performance should be compared under conditions that are as similar as practicable.

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8. Further Development Plan

PRIORITY	ACTIVITY
1	Document the current UI/UX version and graphical prototypes
2	Complete the foundational infrastructure of the team's independent software
3	Define data contracts and software interfaces
4	Progressively connect the interface to real data sources
5	Develop and test mission logic in simulation
6	Implement swarming incrementally in the independent software
7	Develop and evaluate computer-vision models using valid data
8	Investigate operational integration with the flight platform
9	Assess the feasibility of building physical payload prototypes
10	Conduct staged testing and record performance evidence

Capability development must be accompanied by version control, change records, and test reports so that the relationship between software versions and evaluation results remains traceable.

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9. Conclusion

The **Shahid Mohajer** team's project aims to design and develop a coordinated three-drone system for intelligent agricultural protection.

Software development follows two parallel tracks: **UI/UX enhancement of the default skyris.pro platform** and **development of a basic, independent application by the team during the competition**. The first track focuses on improving user interaction, while the second provides a foundation for the progressive implementation and evaluation of swarming.

At present, modular payloads remain at the concept and graphical-prototype stage. Their physical implementation will be considered as part of future development.

The project's central objective is to integrate fleet management, mission information, and analytical capabilities into a coherent workflow. Further development will focus on completing the software infrastructure, conducting staged testing, and documenting performance evaluation.

References

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2. **Delalieux et al., 2023 — Red Palm Weevil Detection in Date Palm Using Temporal UAV Imagery**

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4. **FAO — Plant Production and Protection**

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5. **QGroundControl**

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6. **Mission Planner — Swarming**

https://ardupilot.org/planner/docs/swarming.html

7. **ROS 2 — Quality of Service**

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8. **MAVLink — Message Signing**

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